

KSB Etanorm 100-080 Centrifugal Pump

Operation, Maintenance & Troubleshooting Manual

Applicable Units: P-05, P-06, P-09, P-12 (Diesel / Fuel Oil Service)

1. Pump Specifications

Parameter	Value
Model	Etanorm 100-080-200
Type	Single-stage, end-suction, back pull-out
Service fluid	Diesel / fuel oil / light distillates
Design flow	100 m ³ /hr
Design head	70 m
Rated power	30 kW
Rated speed	2,950 RPM
Impeller diameter	262 mm
Impeller type	Closed, 6-vane
Casing material	EN-GJL-250 cast iron (diesel service)
Impeller material	EN-GJS-400-15 ductile iron
Shaft material	1.4021 martensitic stainless steel
Bearing type	NSK 6310 (drive end), NSK 6306 (non-drive end)
Seal type	Mechanical single seal, API Plan 11
NPSH required	3.0 m at design flow
Maximum allowable temperature	140°C
Maximum allowable pressure	12.5 bar

2. Normal Operating Envelope

2.1 Flow & Head Envelope

- Minimum continuous flow: 40 m³/hr (40% of BEP)
- Best Efficiency Point (BEP): 100 m³/hr at 70 m head
- Maximum continuous flow: 115 m³/hr (115% of BEP)
- Pump efficiency at BEP: 78%

2.2 Bearing Temperature Limits

Condition	Bearing Temp (°C)	Action
Normal operation	45–65°C	No action required
Elevated — monitor	65–78°C	Check lubrication; increase monitoring
Warning threshold	78–85°C	Inspect within 4 hours
Alarm threshold	>85°C	Reduce load; inspect immediately
Failure threshold	>95°C	Immediate shutdown

2.3 Vibration Limits (ISO 10816-3, Group 2)

Zone	Vibration (mm/s RMS)	Condition	Action
A	0 – 2.3	New machine	Normal
B	2.3 – 4.5	Acceptable long-term	Monitor
C	4.5 – 7.1	Unsatisfactory	Plan maintenance within 2 weeks
D	> 7.1	Dangerous	Shutdown immediately

Misalignment signature: Elevated axial vibration (>2× radial) combined with 2× running speed frequency in spectrum — indicates angular misalignment. Realign within 48 hours.

2.4 Back Pull-Out Feature

The Etanorm 100-080 uses a back pull-out design — the rotating assembly (impeller, shaft, bearings, seal) can be removed without disturbing the pump casing or pipe connections. This reduces maintenance downtime significantly: - Typical bearing/seal replacement time: 2–3 hours (vs. 6–8 hours for conventional design) - Casing remains bolted to baseplate and piping during maintenance

3. Bearing Maintenance

3.1 Lubrication Specifications

- Lubricant: NSK Grease LG2 (NLGI Grade 2) or equivalent
- Alternative: Castrol Tribol 1100/220
- Relubrication interval: every 3,000 operating hours
- Grease quantity: drive end 18 g, non-drive end 10 g
- Grease nipple location: bearing housing top (both ends)

3.2 NSK 6310 Bearing Data

- Dynamic load rating (C): 61.8 kN
- Static load rating (C0): 38.0 kN
- Limiting speed (grease): 6,300 RPM
- L10 life at design conditions ($P \approx 6.0$ kN, $n = 2,950$ RPM): $\sim 44,000$ hours

3.3 Bearing Replacement Procedure (Back Pull-Out)

1. Close suction and discharge isolation valves; drain pump casing
2. Disconnect coupling (do not disturb motor or piping)
3. Remove back pull-out unit: loosen $4 \times M16$ casing bolts; slide rotating assembly rearward
4. Place rotating assembly on maintenance bench
5. Remove bearing housing end covers ($6 \times M10$ bolts, torque 25 N·m)
6. Extract bearings using hydraulic puller — support shaft to avoid bending
7. Clean shaft journals; inspect for wear (replace if diameter reduced >0.02 mm)
8. Heat new NSK 6310 to 80°C ; press onto shaft
9. Repack: 18 g grease in drive end housing, 10 g in non-drive end
10. Reassemble back pull-out unit; slide into casing; torque casing bolts to 85 N·m
11. Align coupling: ≤ 0.05 mm parallel, ≤ 0.05 mm angular TIR
12. Run-in: 1 hour at 50% load; verify temperature and vibration

3.4 Misalignment — Common Root Cause for P-05, P-06, P-09, P-12

These units are mounted on a shared baseplate with the motor. Thermal growth of the motor during operation causes misalignment if cold alignment is not corrected for thermal offset: - Cold alignment target: 0.10–0.15 mm low (pump shaft lower than motor shaft) - This compensates for motor thermal growth at operating temperature - Check alignment after first 100 hours of operation on new installations

4. Mechanical Seal Maintenance

4.1 Seal System — API Plan 11

- Flush from pump discharge via 3 mm orifice to seal chamber
- Seal face materials: Silicon carbide (rotating) vs. Carbon graphite (stationary)

- O-ring material: FKM (Viton) — compatible with diesel and fuel oil

4.2 Seal Failure Indicators

- Visible fuel oil leakage at seal housing: shutdown required (fire hazard)
- Seal leak sensor activated: shutdown within 10 minutes
- Flush line temperature $>100^{\circ}\text{C}$: check flush flow; inspect orifice for plugging
- Fuel oil in bearing housing: seal failure — immediate shutdown

4.3 Seal Replacement Procedure (Back Pull-Out)

1. Remove back pull-out unit (see Section 3.3, steps 1–3)
2. Remove gland plate ($4 \times \text{M10}$ bolts)
3. Slide seal off shaft; inspect faces for scratches or chips
4. Clean shaft sleeve; inspect for wear (replace sleeve if worn $>0.1 \text{ mm}$)
5. Install new seal cartridge; verify O-ring seating
6. Torque gland plate bolts evenly: $30 \text{ N}\cdot\text{m}$
7. Reinstall back pull-out unit
8. Typical replacement time: 2–3 hours

4.4 Seal Life Expectancy

- Normal diesel service: 24–36 months
 - Reduced life factors: particulate contamination, temperature $>120^{\circ}\text{C}$, shaft runout $>0.08 \text{ mm}$, dry running
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5. Impeller & Wear Ring Maintenance

5.1 Wear Ring Clearance

- New clearance: $0.30\text{--}0.40 \text{ mm}$ (radial, each side)
- Maximum allowable: 0.75 mm
- Measurement method: feeler gauge through inspection port or dial indicator on disassembled pump

5.2 Impeller Replacement Criteria

- Visible erosion on vane leading edges $>3 \text{ mm}$ deep
- Cavitation pitting covering $>20\%$ of suction vane surface
- Differential pressure decline $>15\%$ from baseline at same flow and speed
- Cracks or fractures (any size) — replace immediately

5.3 Impeller Removal

- Impeller nut: right-hand thread, $\text{M30} \times 2.0$
 - Torque for installation: $120 \text{ N}\cdot\text{m}$
 - Use shaft-locking tool (KSB part 50.10) to prevent shaft rotation during removal
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6. Troubleshooting Guide

Symptom	Probable Cause	Corrective Action
Bearing temp >78°C	Insufficient lubrication	Re-grease per schedule
Bearing temp >78°C	Misalignment	Check cold alignment; correct thermal offset
Axial vibration 2× radial	Angular misalignment	Realign coupling
Vibration at 2× RPM frequency	Misalignment or impeller imbalance	Align; inspect impeller
Flow declining >10%	Wear ring clearance excessive	Measure clearance; replace rings
Seal leak	Seal face damage or dry run	Replace seal; check flush flow
Noise — cavitation	Low NPSH	Check suction line; reduce flow
Motor overcurrent	Impeller rubbing or fluid density change	Check clearance; verify fluid properties
Pump won't prime	Air in suction line	Vent suction; check foot valve

7. Spare Parts Recommendation

7.1 Recommended On-Site Spares (per pump)

Part	Quantity	Reason
Mechanical seal cartridge	1	High wear item
NSK 6310 bearing	2	Drive end — most loaded
NSK 6306 bearing	1	Non-drive end
Wear ring set (casing + impeller)	1	Erosion wear
Shaft sleeve	1	Seal area wear
O-ring set (full pump)	2	Routine replacement
Impeller	1	Cavitation damage risk

7.2 Recommended Fleet Spares (for 4-pump fleet P-05/06/09/12)

- 1 complete back pull-out rotating assembly (ready to swap in <1 hour)
- 2 mechanical seal cartridges
- 4 bearing sets